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Equity Volatility Forecasting: Robustness Across Time and Companies

Matched-information baselines and controls for financial-data availability

In this sample, model rankings vary by year and financial ratios add no consistent benefit.

01 Research question

Do nonlinear models outperform linear forecasts when they receive the **same market history**? Do financial ratios add information beyond missing-data patterns and filing age?

02 Data and prediction target

30 US non-financial large-cap companies. Daily adjusted prices, volume and SEC financial statements. Forecast origins span 2015-2025, with 2014 for feature warm-up.

Inputs: 60 trading days of market history and the latest available financial snapshot. Target: annualized volatility over the next 20 trading days.

$$y_{i,t} = \sqrt{\frac{252}{20}\sum_{k=1}^{20}(r_{i,t+k} - \bar{r}_{i,t+1:t+20})^2}$$

Daily log returns, population standard deviation. This measures a daily-return volatility proxy, rather than intraday realized variance.

03 Experimental design

Matched information: sequence Ridge and LSTM use the same 60 x 10 market observations. Boosting and snapshot Ridge use market summaries. HAR/ML literature motivates these benchmarks [1,2].

Financial ablations: market only, market plus coverage controls, and market plus full financials. Coverage controls retain missing flags and filing age but remove accounting values.

Protocols. Static 2024-2025, annual evaluations in 2020/2022/2024, and a company holdout. Each annual run validates on the previous year and trains on earlier data.

Leakage controls. Training-only preprocessing, prior-date SEC filings, purged overlapping labels and validation-only tuning. LSTM ensembles average three fixed seeds.

04 Model advantages depend on the year

Selected expanding-window evaluations. Both models receive the same market sequence.

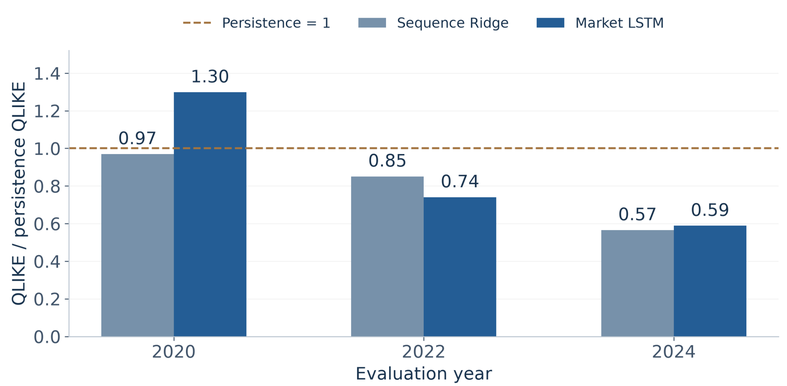


Figure 1. Loss relative to persistence within each year. Lower is better. LSTM beats Ridge in 2022, but not in 2020 or 2024 by point estimate.

05 Transfer to excluded companies

Train and validate on 22 firms. Evaluate 2024-2025 on 8 excluded firms, using the same fitting rules.

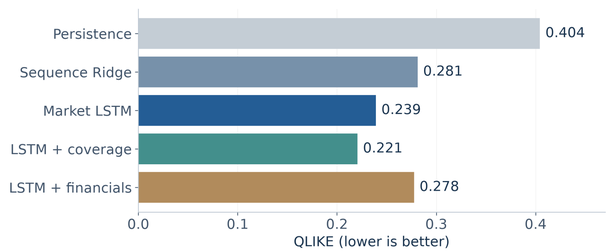


Figure 2. Financial ratios worsen LSTM QLIKE from 0.239 to 0.278 in this cohort.

06 Accounting values versus coverage

The coverage-control LSTM keeps the architecture fixed while removing the seven financial ratios.

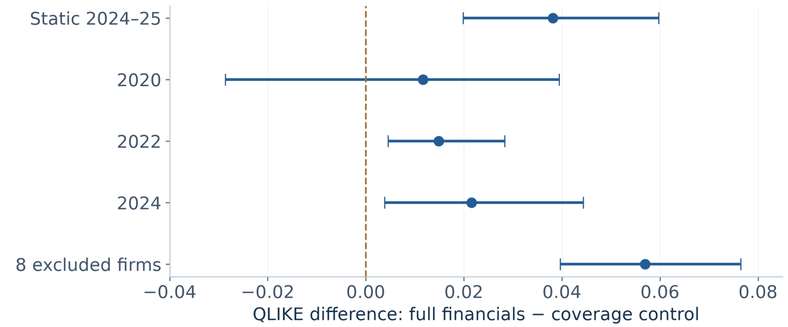


Figure 3. Full financials minus coverage control. Positive values favor the control. Bars show pointwise 95% paired bootstrap intervals using 20-day date blocks. The protocols overlap and are not independent replications.

07 Forecast error and screening differ

Static 2024-2025	QLIKE	Top-5 overlap
Persistence	0.541	45.7%
Sequence Ridge	0.383	58.7%
Market LSTM	0.365	54.4%
LSTM + coverage	0.344	57.0%
LSTM + financials	0.382	56.4%

Screening selects five of 30 companies each day. Lower forecast loss need not produce a better risk ranking.

Conclusion. Strong linear baselines and data-availability controls materially change the interpretation of model gains.

Scope and limitations. V2 protocols follow inspection of V1 results and are exploratory. The 8 excluded firms appeared in V1. The convenience sample has survivorship bias. Mean total-liability and operating-margin coverage is 61% and 66%. Results concern this feature set. Top-5 scores from the 8-firm cohort are not comparable with the 30-firm scores. No investment-return claim.

References. [1] Corsi (2009), J. Financial Econometrics 7, 174-196. [2] Christensen, Siggaard & Veliyev (2023), J. Financial Econometrics 21, 1680-1727. SEC EDGAR Company Facts.

Reproducibility. 5 protocols, 13 specifications, 3 LSTM seeds. Saved configurations, predictions and date-block intervals. AI coding assistance disclosed.